

Offline Facial Recognition and Liveness Detection for Low-Connectivity Field Operations

Prepared for the **NHAI Innovation Hackathon 7.0**

Team

Runtime_Hustle

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Problem Statement

Remote Operations

NHAI field personnel work in **zero-network zones** — cloud authentication fails entirely, leaving critical identity verification impossible in the field.

Identity Fraud

High risk via **printed photo spoofing** or digital replay attacks — traditional systems cannot distinguish a live person from a photograph or screen recording.

Engineering Challenge

Secure authentication must run on **mid-range Android 8.0+ / iOS 12+** devices with an ultra-low model footprint — no high-end hardware assumptions.

Our Solution



100% Offline

Edge face recognition and liveness verification — zero cloud dependencies at any stage of the pipeline.



Dual-Layer Defense

Passive liveness analysis combined with automatic active challenge — blink, smile, or head turn when confidence is low.



Ultra-Compact Footprint

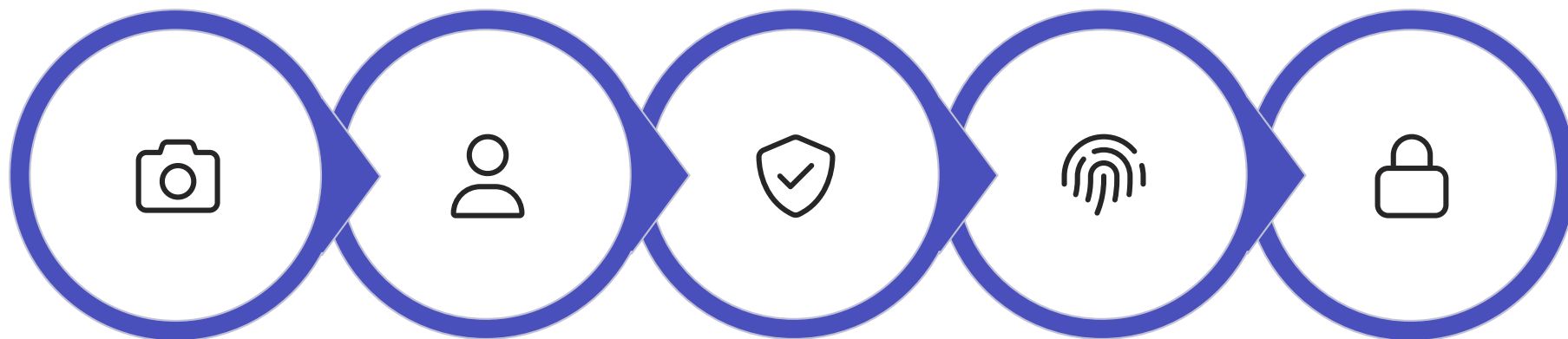
Entire ONNX model stack under **20 MB** — optimized for constrained device storage and memory.



React Native Ready

Designed for frictionless React Native native-module deployment across both Android and iOS platforms.

System Architecture



Capture

Face Detect

Liveness

Embed

Local Match

On-Device Inference

All inference runs locally on device CPU — **zero cloud dependencies** at every stage of the pipeline.

Secure Local Storage

Embeddings matched securely against **local encrypted storage** — no biometric data leaves the device.

Queue & Sync

Attendance logs **queue safely** in a pending_sync store until network connectivity is restored.

Model Stack & Size

Component	Model	Role
Detector	YuNet ONNX	Lightweight face detection
Recognizer	SFace INT8 Quantized ONNX	Accurate recognition, Indian demographics
Liveness	MiniFASNetV2 ONNX	Real-time anti-spoofing



 **Size Target: PASS**

NHAI Requirement: ≤ 20 MB

Our Footprint: **10.99 MB** — 50% under the limit

INT8 quantization applied to SFace delivers **reduced model size with zero accuracy loss**, making it ideal for deployment on mid-range Android and iOS hardware without sacrificing recognition quality for Indian demographic profiles.

Liveness & Anti-Spoofing

Layer 1 — Passive Analysis

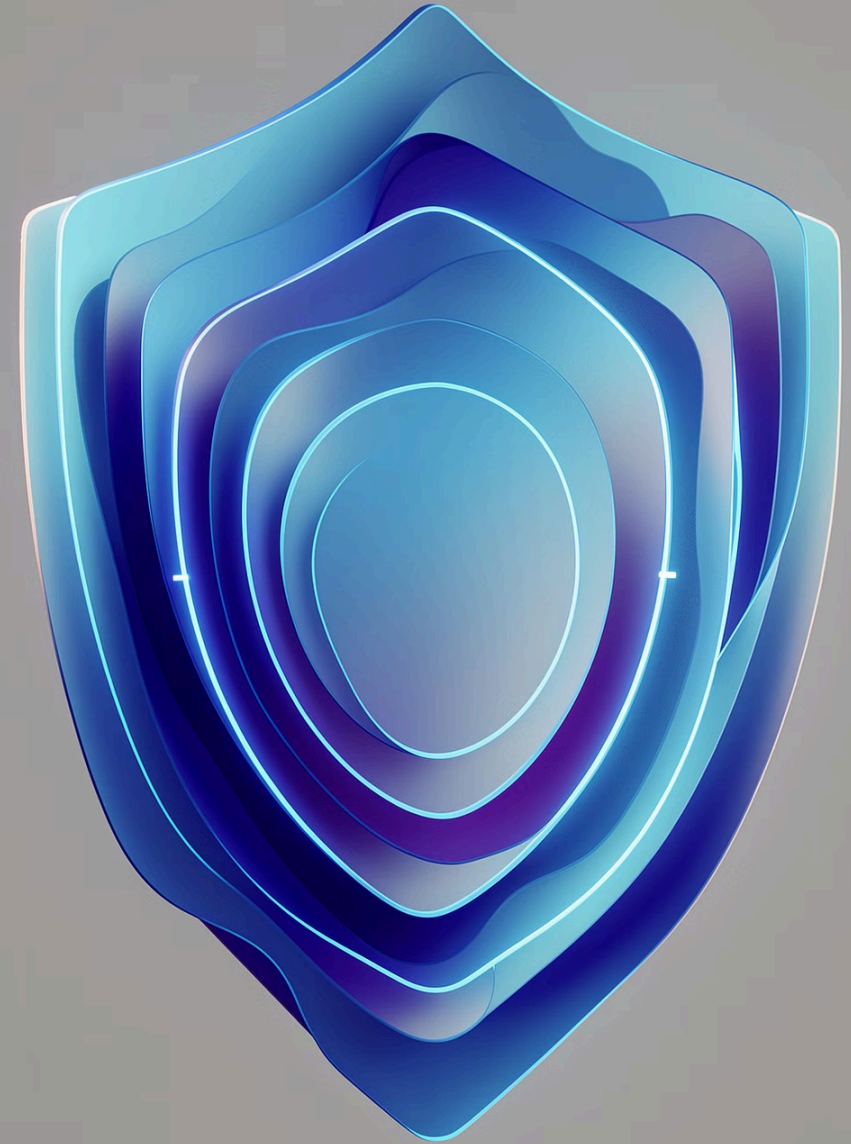
Evaluates **reflections, depth anomalies, and structural textures** on every captured frame. Runs silently in the background with no user interaction required — completely frictionless for genuine users.

Layer 2 — Active Challenge

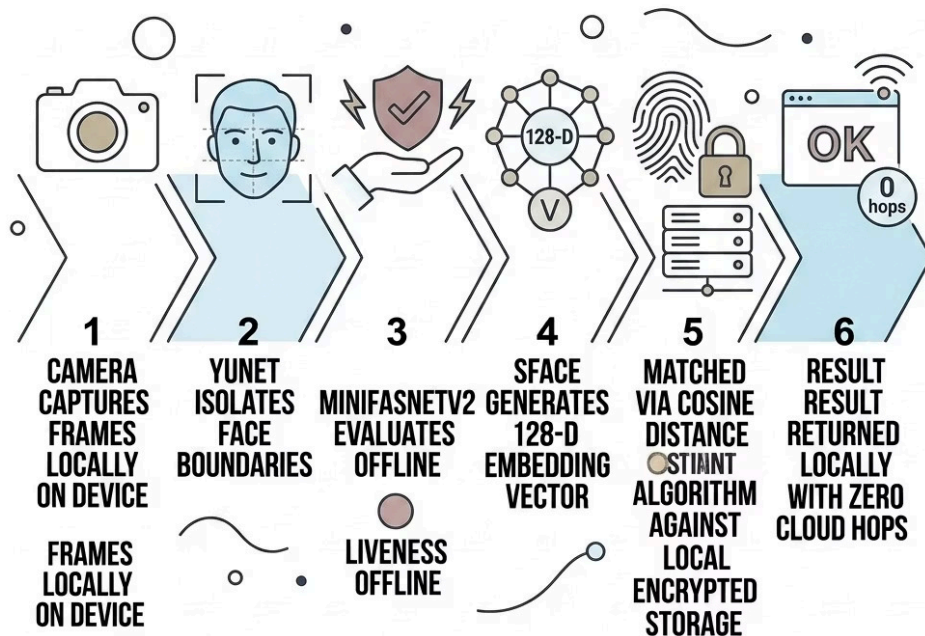
UI prompts the user to **blink, smile, or perform a head turn** when passive confidence falls below threshold. Dynamically triggered only when needed to preserve user experience.

Result — Complete Spoofing Defense

Blocks **printed photo clones and digital replay attacks** while maintaining a frictionless UX for legitimate field personnel. No cloud call required at any point.



Offline Verification Flow



Zero Cloud Hops

Every step of the verification pipeline executes entirely on the device. From frame capture to final authentication result, **no data is transmitted over any network**.

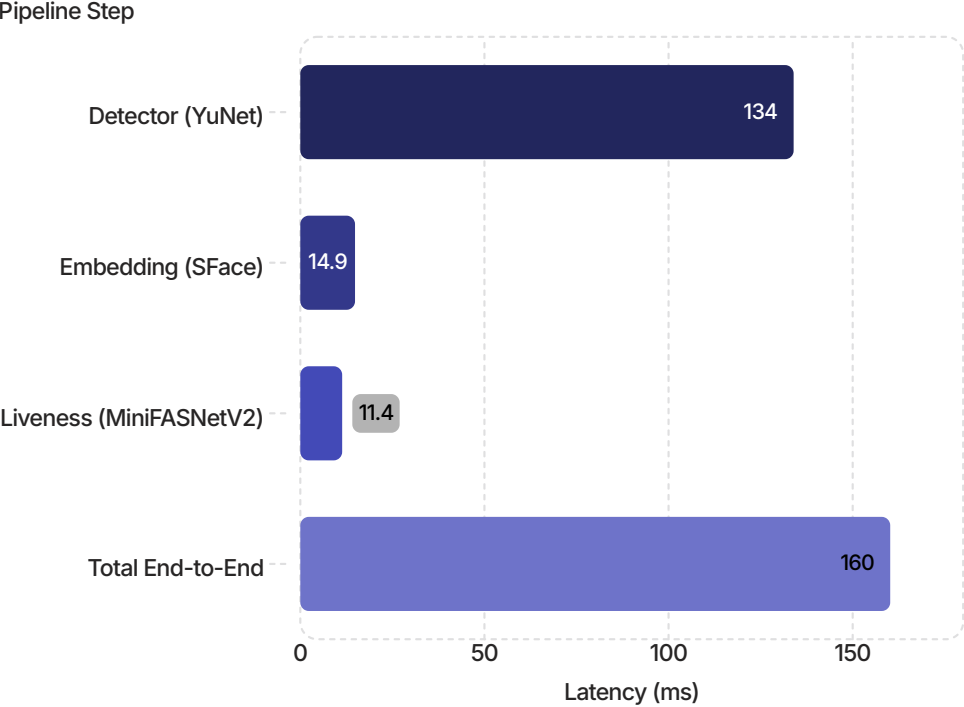


128-D Embedding Vector

SFace produces a compact 128-dimensional facial embedding that uniquely identifies an individual while remaining computationally efficient for on-device cosine distance matching.

The local match uses a **distance threshold algorithm** to compare the live embedding against enrolled templates stored in encrypted local storage — delivering a pass/fail result in milliseconds.

Performance Benchmarks



✓ Latency Target: PASS

NHAI Requirement: < 1,000 ms

Our Result: ~160.19 ms — 84% faster than target

Benchmark Breakdown

Step	Latency
Detector (YuNet)	133.96 ms
Embedding (SFace)	14.86 ms
Liveness (MiniFASNetV2)	11.37 ms
Total End-to-End	~160.19 ms ✓

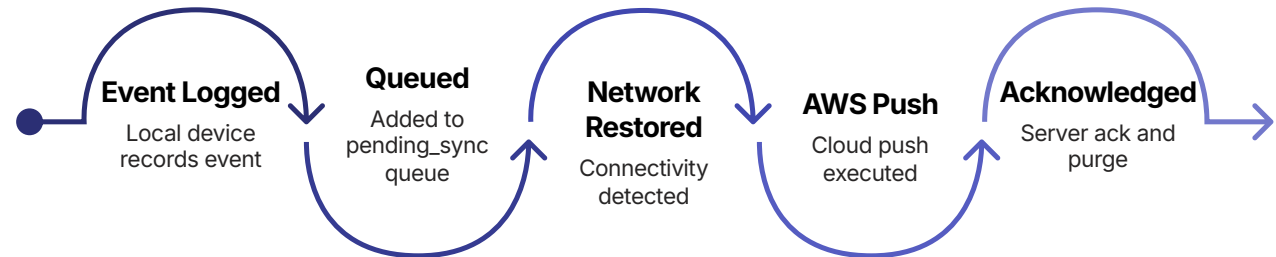
All benchmarks measured on mid-range Android hardware — representative of NHAI field device profiles.

React Native Integration & Sync

Sync Lifecycle

Integration Architecture

Validated via **Python prototype** before production deployment. ONNX models are deployed via native ONNX Runtime Mobile bridges for both Android and iOS.



Public API Surface

Attendance records accumulate safely in a local `pending_sync` queue during offline operation. When connectivity is restored, records are pushed to AWS, acknowledged, and then purged from the device — ensuring data integrity with no duplication.

initializeModels()

Loads ONNX model stack into memory on app launch

registerFace()

Enrolls a new user embedding into local encrypted storage

verifyFace()

Runs full liveness + recognition pipeline and returns result

purgeSyncedRecords()

Clears acknowledged records from local queue after cloud push

Validation, Innovation & Next Steps

Validation Snapshot

✓ Indoor/Outdoor Lighting

Verified across varied ambient lighting conditions representative of NHA1 field environments.

🔒 Printed Photo Attack Blocked

Dual-layer liveness successfully rejects all printed photograph spoofing attempts.

🔒 Screen Replay Blocked

Digital replay attacks via screen playback are detected and rejected by passive analysis.

✓ Model: 10.99 MB | Latency: ~160.19 ms

Both NHA1 size and speed targets met with significant margin.

🏆 Innovation Highlights

50% under the 20 MB size limit · Sub-second speed at ~160 ms · Fully offline dual-layer anti-spoofing — no comparable open solution exists at this footprint.

Next Steps

01

Native Bridge Compilation

Compile production-grade C++/Java/ObjC native bridges for full React Native deployment

02

Low-Tier Chipset Stress Testing

Validate performance on entry-level Snapdragon and MediaTek chipsets used in field devices

03

Field Condition Validation

Shadow, glare, and extreme outdoor lighting validation in real NHA1 site environments